ANANT GOEL

IoT / Robotics Software Intern

anantgoel2005@gmail.com | India

builtbyanant.site | github.com/Anantgoel2005 | linkedin.com/in/ag25goel

SUMMARY

Final-year Computer Science student with an electronics and IoT internship. Building connected-device APIs, sensor telemetry, fog-edge simulations, and perception workflows.

EDUCATION & AVAILABILITY

BML Munjal University | B.Tech in Computer Science | Expected May 2027

Seeking on-site / hybrid internships in India. Flexible duration; start date to be discussed.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript

IoT software: Device REST APIs, RFID workflows, sensor logging, SQLite

Perception & simulation: OpenCV, YOLO, ByteTrack, calibrated geometry, virtual sensor nodes

Tools: Node.js, Express, Flask, Git, automated tests

WORK EXPERIENCE

Intern | Bodycast Innovators Pvt. Ltd. | May - July 2025

Dwarka, Delhi, India | On-site

- Assisted across multiple projects while developing practical familiarity with power electronics, IoT, and microcontrollers.
- Gained exposure to PCB design, consumer electronics, and electrical engineering through the internship.

SELECTED PROJECTS

RFID Smart Warehouse System | JavaScript, Node.js, Express, SQLite

- Built REST APIs for RFID registration, tag assignment, checkout, and environmental sensor telemetry, with SQLite audit logs and a dashboard.
- Designed device integration for ESP32, RC522, DHT22, and MQ sensors; the public repository contains the server and dashboard.

SpeedNet | Fog-Edge Simulation | Python, Flask, JSON telemetry

- Built a Flask fog-layer API and deterministic virtual sensor nodes for explainable weather- and traffic-based speed advisories.
- Added validated JSON ingestion, stale-node expiry, thread-safe state, a live dashboard, and automated tests; inputs are simulated.

PathWise | Python, YOLO, ByteTrack, OpenCV

- Built vehicle tracking, calibrated road-plane projection, and explainable time-to-collision and cut-in rules using YOLO, ByteTrack, and OpenCV.
- Validated core risk logic with a reproducible 180-frame synthetic scenario, covering 540 actor samples.